**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

MGU–UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 Admission Onwards)
**Course Code: MG4DSEECT205**

**Course Title: Continuous and Discrete Systems**

**Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability***

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

**Duration: 1½ Hours** **Maximum Marks: 50**

## Section A – Short Answer
**Answer any TEN questions**
***(10 × 2 = 20 Marks)***

1. Define the operating point (Q-point) of a BJT. **U 1**
2. Mention any two modes of operation of a BJT. **K 1**
3. What is voltage divider biasing? **U 1**
4. Define an RC coupled amplifier. **K 1**
5. State the Barkhausen criterion for oscillations. **K 1**
6. What is meant by slew rate of an operational amplifier? **K 2**
7. Define Common Mode Rejection Ratio (CMRR). **K 2**
8. What is an inverting amplifier? **U 2**
9. What is the function of an integrator circuit? **U 2**
10. Define binary number system. **K 3**
11. Find the 1's complement of the binary number (10110101)2 **A 3**
12. What is a half adder? **K 3**
13. What is a flip-flop? **K 3**
14. What is a ring counter? **U 3**

## Section B – Short Essay
**Answer any SIX questions**
*(6 × 5 = 30 Marks)*

15. Explain the operating point of a BJT and discuss the modes of operation with suitable diagrams. **U 1**

16. Describe the voltage divider biasing of a BJT amplifier and explain its advantages. **An 1**

17. Explain the principle and working of an RC phase shift oscillator. **U 1**

18. With a neat block diagram, explain the internal structure of IC 741 operational amplifier and discuss its important parameters. **U 2**

19. Explain inverting and non-inverting op-amp amplifiers with circuit diagrams and expressions for voltage gain. **A 2**

20. Explain op-amp based integrator and differentiator circuits with diagrams and applications. **U 2**

21. Explain active filters using operational amplifiers (low pass and high pass) and mention ideal and practical characteristics. **An 2**

22. Explain the working of half adder and full adder with logic expressions and truth tables. **A 3**